

# Braillix - Braille Readability Review

Would a blind person actually be able to read this? Reviewed 2026-07-29 against Library of Congress / Marburg Medium braille specifications. All "Braillix" numbers taken from `cad/scad/mech_layout.scad`, not from memory.

## 1. Scorecard

| Parameter                             | Braille standard | Braillix | Verdict                 |
|---------------------------------------|------------------|----------|-------------------------|
| Dot base diameter                     | 1.44 - 1.60 mm   | 1.5 mm   | [OK] IN SPEC            |
| Dot height (travel)                   | 0.46 - 0.50 mm   | 0.8 mm   | [!] 1.6x too tall       |
| Within-cell <b>vertical</b> spacing   | 2.30 - 2.50 mm   | 2.6 mm   | [OK] marginal (4% over) |
| Within-cell <b>horizontal</b> spacing | 2.30 - 2.50 mm   | 4.8 mm   | [X] 1.9x too wide       |
| Cell-to-cell pitch                    | 6.10 - 6.50 mm   | 68 mm    | [X] 10.5x too wide      |

**Two of five are correct, one is marginal, two are not.** The two failures are different in kind: one is fixable, one is architectural.

## 2. What we're getting RIGHT

**The dot itself is now genuinely correct.** At 1.5 mm base diameter it sits squarely inside the 1.44-1.60 mm standard. A trained reader's fingertip would register it as a proper braille dot, not a lump.

This happened almost by accident - the dot shrank from 2.2 mm to 1.5 mm in v7.1 because a 2 mm spring had to fit around it. A constraint from the mechanism pushed the design *toward* the standard rather than away from it. Worth saying out loud in the presentation: the dot is spec-compliant.

**Vertical spacing is effectively correct** at 2.6 mm against a 2.30-2.50 mm standard. 4% over is inside the tolerance a reader would notice.

**The dot is a smooth printed dome**, not a glued-on bearing ball with an adhesive fillet around its base. Readers detect dot *shape*, and a clean hemispherical dome is the right profile.

## 3. What needs attention

### [!] Dot height 0.8 mm - 1.6x the standard, and easy to fix

Standard braille dots rise **0.46-0.50 mm**. Ours rise **0.8 mm**. A dot this proud feels sharp and "spiky" to an experienced reader, and it makes a raised dot harder to distinguish from the plate edge.

**This is the single most fixable readability problem.** It is one parameter - `pin_lift` in `mech_layout.scad` - and lowering it to 0.5 mm would:

- bring dot height into spec, **and**
- ease the cam ramp problem at the same time (a shorter rise needs a shorter ramp, which buys back flat dwell on the outermost track, which was the tightest constraint in the whole cam)

**Caveat:** changing `pin_lift` ripples through the vertical stack (`link_total_h`, flange clearance, spring working length). It is a ~1 hour change with re-verification, **not** something to do the week of a presentation. Recommend: after the demo.

### [X] The cell is DISTORTED, not "jumbo" - and this may now be fixable

This is the finding worth taking seriously.

|                    |        |                 |
|--------------------|--------|-----------------|
| vertical spacing   | 2.6 mm | (~ standard)    |
| horizontal spacing | 4.8 mm | (1.9x standard) |

A genuine large-format braille cell scales **both** axes together. Ours is standard vertically but nearly double horizontally, so it is not a scaled cell - it is a **stretched** one. It matches no braille standard, and to a trained reader it would feel wrong in a way that "everything is bigger" would not.

The 4.8 mm figure was locked back when the design used **2.5 mm dot holes**: two 2.5 mm holes on a 2.5 mm pitch touch edge-to-edge and merge, so the columns had to be spread.

**That constraint no longer exists.** After v7.1, the holes are **1.7 mm**. At standard 2.5 mm pitch that leaves **0.8 mm of wall** between holes - entirely printable in resin.

So the reason for 4.8 mm was removed by a later change, and nobody revisited it. Bringing columns to ~2.5 mm would make the \*within-cell\* geometry genuinely standard-compliant - a far stronger claim for both the presentation and any future patent filing.

**What it would cost:** `dot_pos()` changes ( $x = +/-1.25$  instead of  $+/-2.4$ ), which recomputes every linkage arm length and assembly angle, plus a re-check that the 2.2 mm spring bores still clear at the tighter pitch (0.3 mm walls - tight, needs verification). Half a day with full verification. **Not before the demo** - but this is the highest-value design improvement remaining in the project.

### **[X] 68 mm between characters - architectural, do not try to "fix"**

A 28BYJ-48 motor is 28 mm in diameter. One motor per character therefore cannot sit at 6.1 mm reading pitch - not with better CAD, not with tighter tolerances. It is geometry, not engineering debt.

**A finger cannot sweep a word.** Standard braille lets one fingertip (~10-12 mm pad) travel continuously along a line; at 68 mm pitch the reader lifts and repositions for every character. That is not reading, it is sequential character identification - a different, much slower cognitive task.

**This is why the honest framing is "modular braille teaching device", not "refreshable braille display".** Own it in the presentation. Evaluators respect a team that knows the limits of its own architecture; they do not respect a team that claims a \$2000 display and demos something 11x too wide.

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## **4. So - could a blind person read it?**

**One character at a time: yes, probably.** The dot size is correct, vertical spacing is correct, and the dome profile is right. A braille reader placing a finger on a single cell would likely identify the character - though the horizontal stretch would feel unfamiliar and slow them down.

**A word or a sentence: no.** At 68 mm per character it is not readable text; it is a sequence of individually-inspected symbols.

**Which is exactly right for the product this actually is.** For a \*learner\* - a child being taught which dots make an "a" - a large, clearly separated, slightly-taller-than-standard dot on a single cell is arguably \*better\* than a compact standard cell. The design's weaknesses as a reading device are close to strengths as a teaching device.

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## **5. Accessibility features worth adding for the demo**

Cheap, fast, and they demonstrate that accessibility was designed in rather than bolted on:

**1 Audio pairing (highest impact, ~\$100).** A small speaker or buzzer that speaks or beeps the character

as it is displayed. Turns a silent mechanism into a multi-sensory teaching tool and gives evaluators something to \*hear\*, not just watch. This is the single best demo addition available.

**1 Tactile orientation markers - already in the design.** The chevron on the box front and the ridge on

the bottom edge tell a blind user which way is up by touch. Point this out; it is exactly the kind of detail evaluators look for.

**1 Tactile part identification - already in the design.** The pod lid has raised ridges, and the nav

caps carry `<|()` `|>` symbols so the buttons are distinguishable without sight.

**1 Homing feedback.** Have the cell audibly or visibly confirm it found home on power-up. Shows the

hall-sensor subsystem working and reassures a blind user that the device is ready.

**1 Say the words "we consulted the standard".** Bring this scorecard. Being able to state

"our dot is 1.5 mm against a 1.44-1.60 mm standard; our character pitch is deliberately non-standard because of the motor, and here is why" is far more convincing than claiming everything is perfect.

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## **6. Recommended actions, in order**

| Priority | Action                                                           | Effort      | When       |
|----------|------------------------------------------------------------------|-------------|------------|
| 1        | Add audio feedback for the demo                                  | ~2 h + ?100 | Before PPT |
| 2        | Bring this scorecard to the presentation                         | 0           | Before PPT |
| 3        | Reduce pin_lift 0.8 -> 0.5 mm (in spec + eases cam ramps)        | ~1 h        | After PPT  |
| 4        | Investigate column pitch 4.8 -> 2.5 mm now that holes are 1.7 mm | ~4 h        | After PPT  |
| 5        | Keep the 68 mm pitch and the teaching-device framing             | 0           | Permanent  |